

STANINZIVO

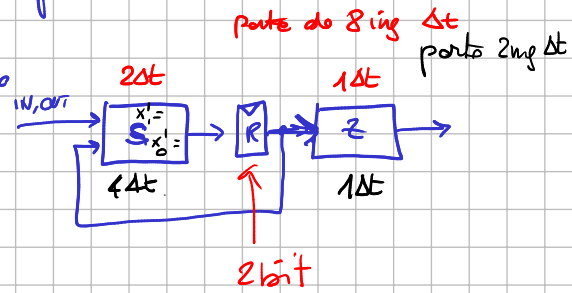
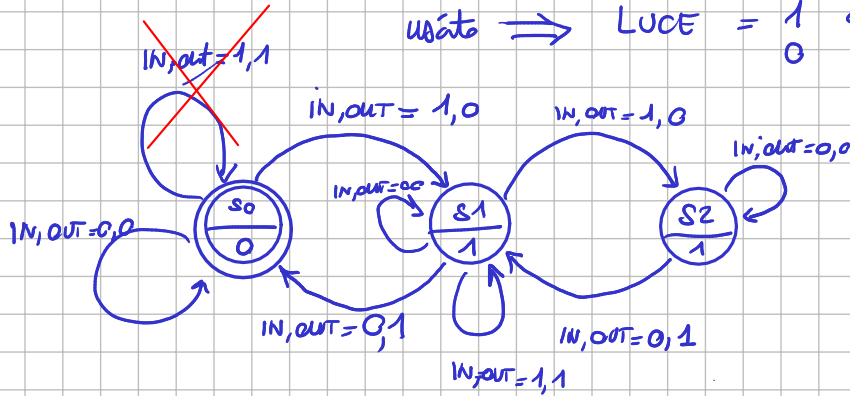
2 persone di copianta



$\left. \begin{matrix} IN \\ OUT \end{matrix} \right\} \begin{matrix} = 1 & \text{entro 1 persona} \\ = 1 & \text{esce 1 persona} \end{matrix}$

0 non succede nulla

usato $\Rightarrow$ LUCE = 1 acceso
0 spento



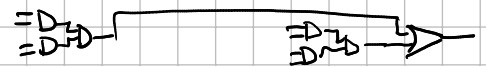
IN 0,1
OUT 0,1

$S_0 = 00$
 $S_1 = 01$
 $S_2 = 10$

$x_1 x_0$

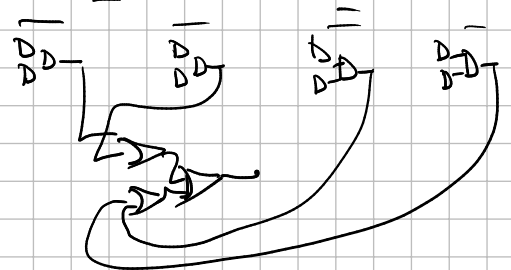
IN	OUT	x_1	x_0	x_1'	x_0'
1	0	0	0	0	1
0	0	0	0	0	0
1	0	0	1	1	0
1	1	0	1	0	1
0	1	0	1	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	1	1	0	0	1

$$Z = X_1 + X_0$$

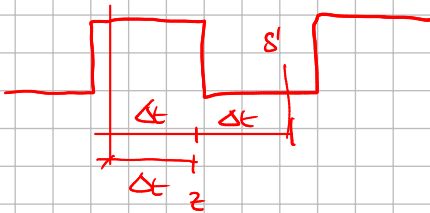


$$X_1' = \overline{IN} \overline{OUT} \overline{x_1} x_0 + \overline{IN} \overline{OUT} x_1 \overline{x_0}$$

$$X_0' = \overline{IN} \overline{OUT} \overline{x_1} \overline{x_0} + \overline{IN} \overline{OUT} \overline{x_1} x_0 + \overline{IN} \overline{OUT} x_1 \overline{x_0} + \overline{IN} \overline{OUT} x_1 x_0$$



porta da 8mg Δt



$$Z = \max\{2\Delta t, 1\Delta t\} + t_{write}$$

$$Z = \max\{4\Delta t, 1\Delta t\} + t_{write}$$

SAVINIZIVO

2 persone di copianta



IN = 1 entro 1 persona
OUT = 1 esce 1 persona

0 non succede nulla

uscita $\Rightarrow$ LUCE = 1 acceso
0

COMPONENTI STANDARD

if (contatore == 0) {
 entro = 0;
 luce spento;
}
if (contatore > 0) {
 esce = 0;
 luce acceso;
}
if (contatore > 1) {
 esce = 1;
 luce spento;
}

Init {
 luce spento;
 contatore = 0;
}
"disp" {
 while (true) {
 if (entro) {
 contatore ++;
 }
 if (esce) {
 contatore --;
 }
 if (contatore == 0) {
 luce spento;
 }
 else {
 luce acceso;
 }
 }
}

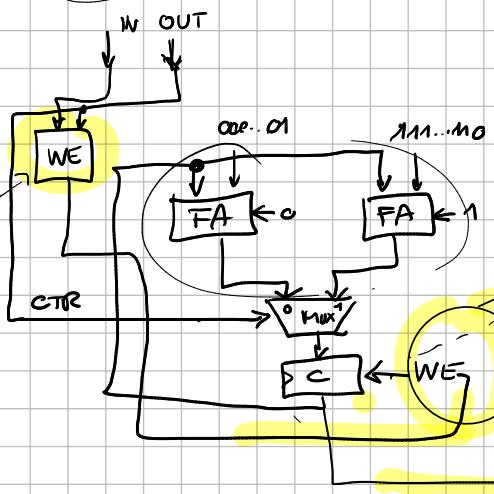
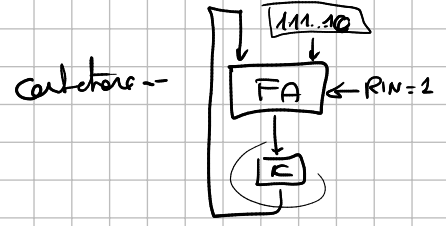
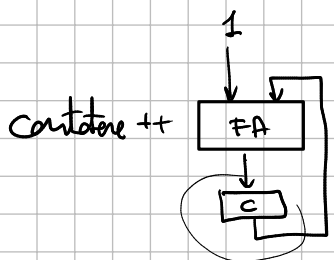
CTR

OUT	0	1
W	-	1
0	-	-
1	0	-

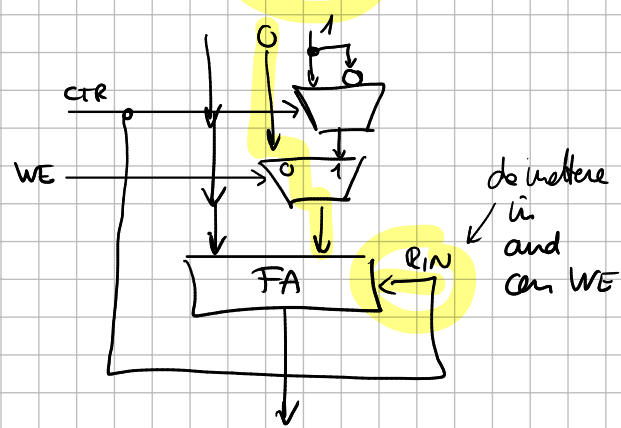
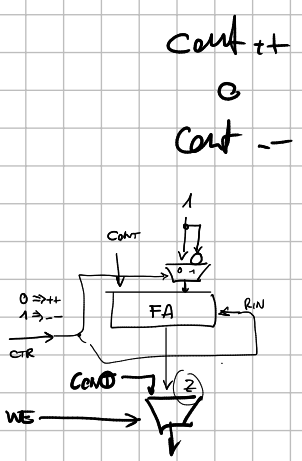
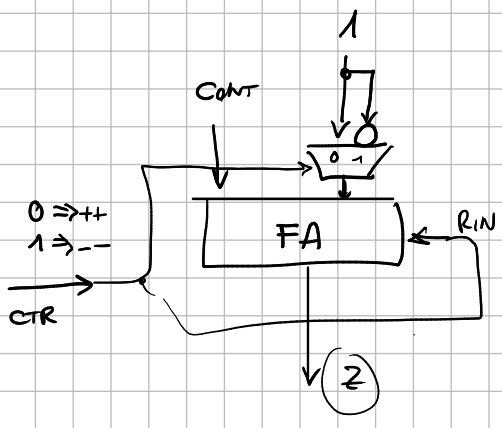
CTR = $\overline{IN} \cdot OUT$
~~CTR = $\overline{IN} \cdot OUT$~~
 CTR = $\overline{IN}$
 CTR = OUT

W	0	1	CTR	WE
1	0	0	0	1
0	1	1	1	1
0	0	0	0	0
1	1	1	0	0

$WE = IN \cdot OUT + \overline{IN} \cdot OUT$



00
01
10



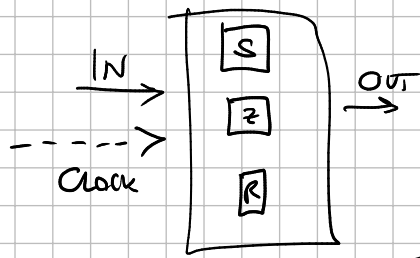
VERILOG

REG

RETN SEQ

component
+ connection

behavioural



```
module m1(output z, input [1:2]x, input clock);
```

```
  wire [1:2]outreg;
```

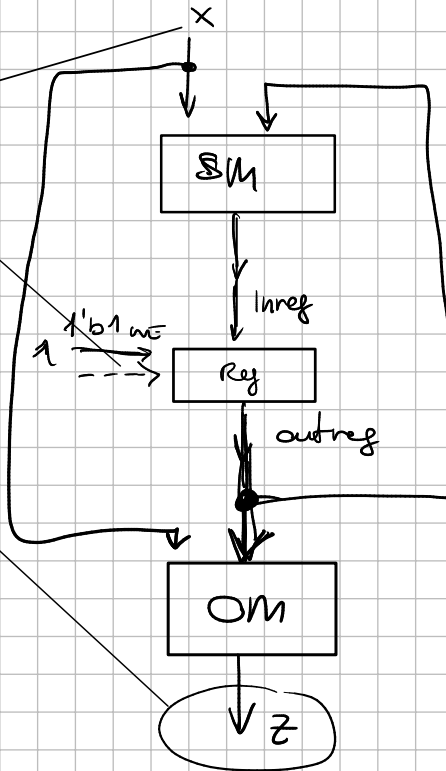
```
  wire [1:2]inreg;
```

```
  register s(outreg, clock, 1'b1, inreg);
```

```
  sigma sm(inreg, outreg, x);
```

```
  omega om(z, outreg, x);
```

```
endmodule
```



Verilog

2 = 1
b = 0

b = 1

c = 1

```
begin
  a = b; 1o
  c = a; 2o
end
```

2 = 0
c = 0

2 1
b 0

```
begin
  a <= b;
  c <= a;
end
```

2 = 0
c = 1

Quartus

